



Gabes University

Summer internship report

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ELITECH CONSULTING

Title:

Integrated HR Management
Platform for Small IT Companies

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Software Engineering and
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Internship Report

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Dedication

To the entire team at Elitech Consulting,

As I conclude my internship as a technician with your company, I would like to express my heartfelt gratitude to each and every one of you.

First and foremost, I extend my thanks to my supervisors and colleagues for your warm welcome, attentive guidance, and the valuable technical insights you have shared with me. Your expertise and patience have played a significant role in my learning and professional growth.

I would also like to thank my teachers and trainers, who equipped me with the theoretical knowledge needed to tackle the technical challenges I faced.

To my friends, for their constant encouragement throughout my journey, and to all those who believed in my potential.

Lastly, I dedicate this internship report to everyone who contributed, directly or indirectly, to this formative experience. Your commitment to my professional success will remain a lasting memory.

With heartfelt gratitude,

Chouikhi Abdallah

Appreciation

First and foremost, we would like to express our deepest gratitude to the board of directors of Elitech for granting us the opportunity to undertake our internship within their company. The welcoming and stimulating environment was instrumental in our growth and development.

A special thanks goes to the technical team, whose remarkable spirit of collaboration and initiative was invaluable. Their support, suggestions, and advice provided critical assistance throughout the project.

We would also like to extend our sincere thanks to our supervisor, whose warm welcome, dedicated time, and daily sharing of expertise were essential pillars of our experience. Her trust in us was a driving force that motivated us to complete our tasks, especially during the more challenging moments when her invaluable guidance steered us towards success.

With heartfelt appreciation,

Chouikhi Abdallah

General Introduction

The opportunity to participate in a project within a company is always valuable for gaining practical knowledge. It is essential to confront new challenges and explore technologies that are of great significance in the field of computer science.

During my first year of studies in Software engineering, I had the chance to undertake a technician internship at Elitech, a company specializing in software development and Trainings.

One of the most exciting aspects of my internship was the project I was assigned: the development of TECH HR, an Integrated HR Management Platform tailored for small IT companies. The system is designed to streamline various HR processes with three key user roles: HR Administrator, who manages human resources, training, and payroll; Employee, who can view tasks, track progress, and update task status; and Manager, who is responsible for assigning and overseeing tasks. TECH HR aims to enhance efficiency in HR management, task distribution, and employee engagement.

In the first chapter, we will focus on the functional requirements specification. This initial step was crucial to understanding the detailed needs of the HR management system for small IT companies.

In the second chapter, we will delve into the conceptual study, examining use case diagrams, sequence diagrams, and class diagrams that helped model and understand the HR management system.

Finally, in the third chapter, we will cover the project implementation, exploring the development environment, programming languages used, and key interfaces of the HR management system. We will also discuss the challenges faced and the valuable lessons learned from this experience.

This internship at Elitech has been an exceptional opportunity, enriching my academic training and significantly preparing me for my future career in computer science.

Chapter 1 : Specification of functional requirements

I. Introduction

In this chapter, we will delve into the key components of the project. We will start with a detailed description, followed by the identification of the involved actors. Finally, we will analyze the functional and non-functional requirements, providing a comprehensive overview of the project.

II. Project description

The main objective of our platform, TECH HR, is to streamline HR processes and enhance task management for small IT companies. The platform aims to simplify human resource management, training, payroll, and task distribution. Each feature is designed to offer a user-friendly experience for HR administrators, managers, and employees alike. TECH HR effectively combines core HR functionalities with an intuitive interface, making it easier for companies to manage employee information, track tasks, and oversee performance. In essence, our platform merges the best practices of HR management with a seamless digital experience, fostering productivity and organizational growth.

III. Actors Identification

On our platform, three actors are involved, each with specific functionalities:

HR Admin:

- Create an account
- Login
- Manage employee data
- Manage training and payroll
- View and update their profile

Manager:

- Create an account
- Login
- Assign tasks to employees
- Monitor task progress
- View and update their profile

Employee (Developer):

- Create an account
- Login
- View assigned tasks
- Update task status (e.g., in progress, completed)
- View and update their profile

IV. Requirements analysis

Let's now explore the project requirements. This will help us understand what the platform needs to achieve in terms of functionality and performance. By analyzing these requirements in detail, we are preparing to implement a successful solution to meet our objectives.

IV.1. Functional Requirements

Our platform allows the HR Admin to:

- **Login to Admin Account:** The HR Admin can log into a secure account using their credentials.
- **Profile Modification:** The HR Admin can update their profile, including security settings.
- **Manage Employee Data:** The HR Admin can add, modify, and delete employee information, including personal details and job roles.
- **Manage Training and Payroll:** The HR Admin can update and manage employee training and payroll information.
- **Manage Leaves:** The HR Admins can manage the Leave Request of the users including the managers and the employees

Our platform allows the Manager to:

- **Login to Manager Account:** The Manager can log into their account using their credentials.
- **Assign Tasks:** The Manager can assign tasks to employees within the same department they manage, including setting deadlines.
- **Monitor Task Progress:** The Manager can track the status of assigned tasks and view updates from employees.
- **View and Update Profile:** The Manager can view and update their personal and professional details.
- **Provide Feedback on Tasks:** The Manager can provide feedback or comments on completed tasks, helping to guide employees.
- **Request a Leave :** The manager can request a leave which will be processed by the HR admin

Our platform allows the Employee (Developer) to:

- **Create an Account:** The Employee can create a user account on the platform.
- **Login to Account:** The Employee can log into their account using their credentials.
- **View Assigned Tasks:** The Employee can view tasks assigned by the Manager.
- **Update Task Status:** The Employee can mark tasks as in progress or completed.
- **View and Update Profile:** The Employee can view and modify their profile, including contact details and work-related information.
- **Access Training Information:** The Employee can view and access available training resources.

IV.2. Non-Functional Requirements

- **Security:** Ensures the protection of users' personal and financial data.
- **Performance:** Guarantees that the platform operates quickly and smoothly, even during high traffic periods.
- **Usability:** Provides an intuitive and user-friendly interface to facilitate navigation and task management.
- **Availability:** Ensures the platform is operational at all times, minimizing downtime.
- **Compatibility:** Ensures the platform works correctly across different browsers and devices (computers, tablets, mobile phones).
- **Error Management:** Provides clear and helpful error messages to assist users in resolving issues.

V. Conclusion

In summary, this first section has guided us through the foundations of the project. We started with an insightful introduction, followed by a detailed description. We also identified the key participants and analyzed the fundamental requirements, both functional and non-functional. With this solid framework in place, we are now ready to dive deeper into the implementation of the project.

Chapter 2: Conceptual Study

I. Introduction

In this chapter, we will explore the key elements of this project through essential diagrams. We will begin with a use case diagram that illustrates the main interactions. Next, we will dive into the details with a sequence diagram, which describes the key steps chronologically. Finally, we will examine the structure of the project through a class diagram, which will visually represent the relationships between the components.

II. Use case Diagram

This diagram will highlight the essential interactions between users and the system, providing a clear perspective on how the project will be used in various scenarios.

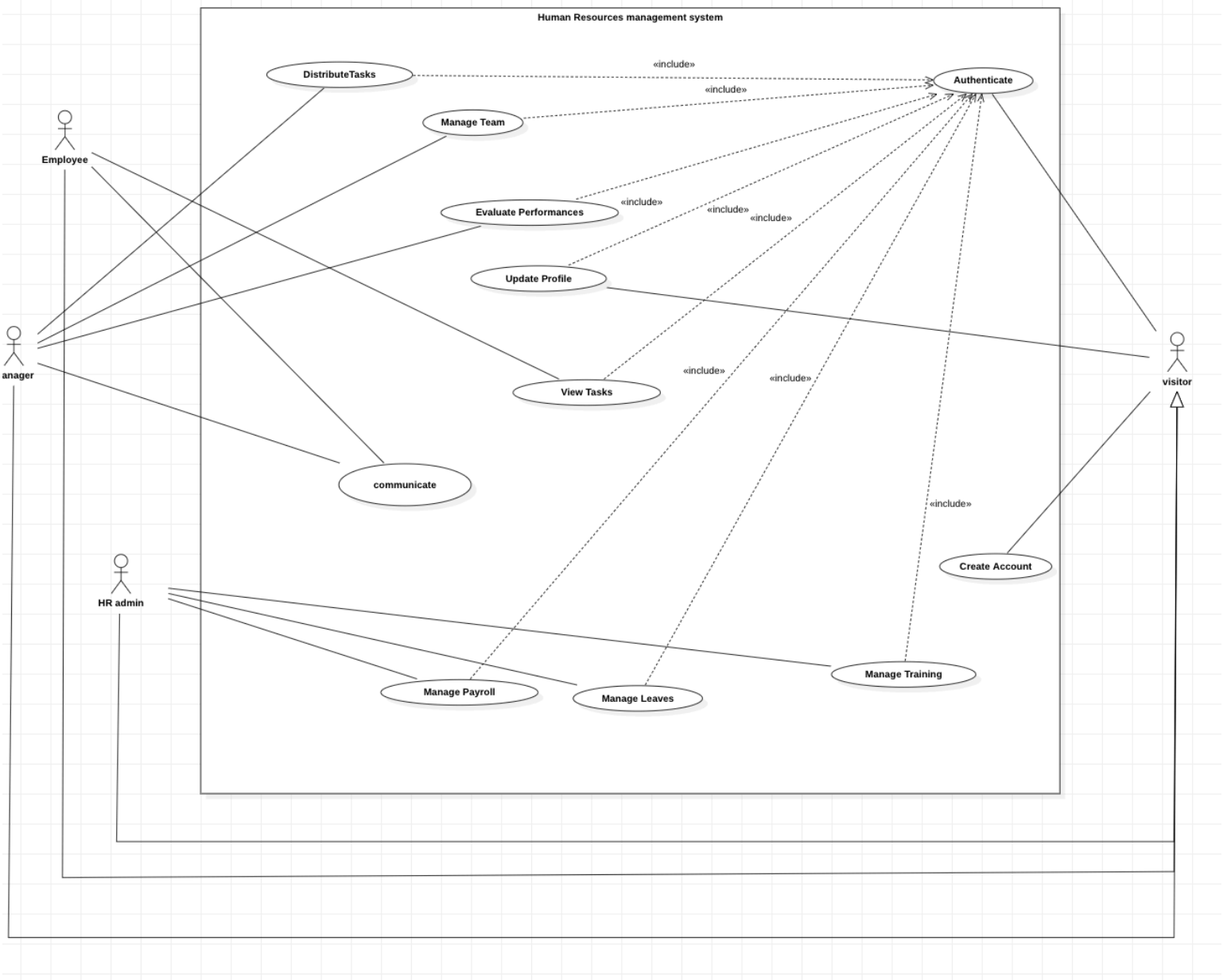


Figure 1. Use Case Diagram

III. Sequence Diagram

This detailed chronological diagram illustrates the sequence of actions between the various components of the system, providing an in-depth view of the key processes. Continuing with our exploration, we will focus on three specific points: the Sequence Diagrams for the Login Process, Forgot Password, and Manager Task Distribution. These diagrams will allow us to visually represent the detailed steps and essential interactions related to these features, as shown in the following figures.

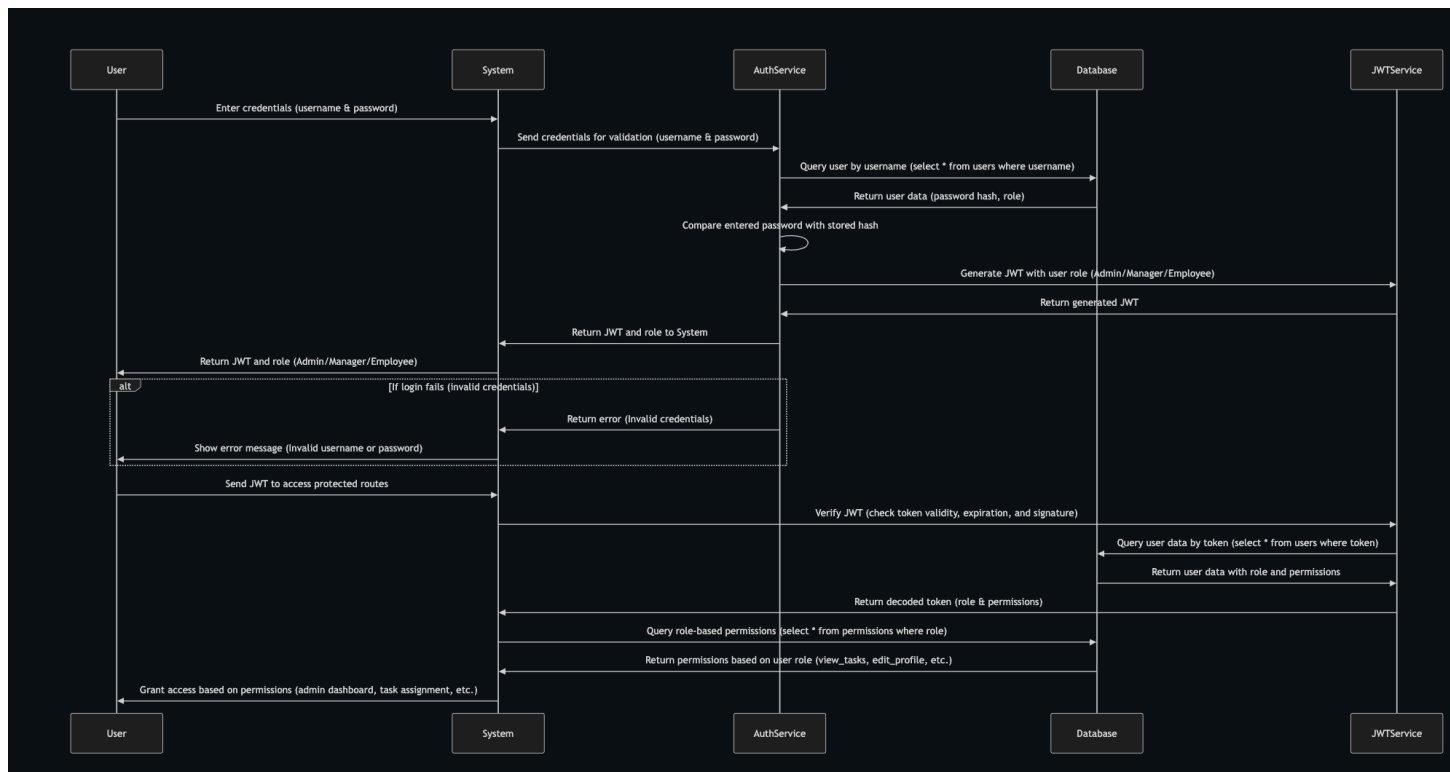


Figure 1. sequence diagram for login process- role-based

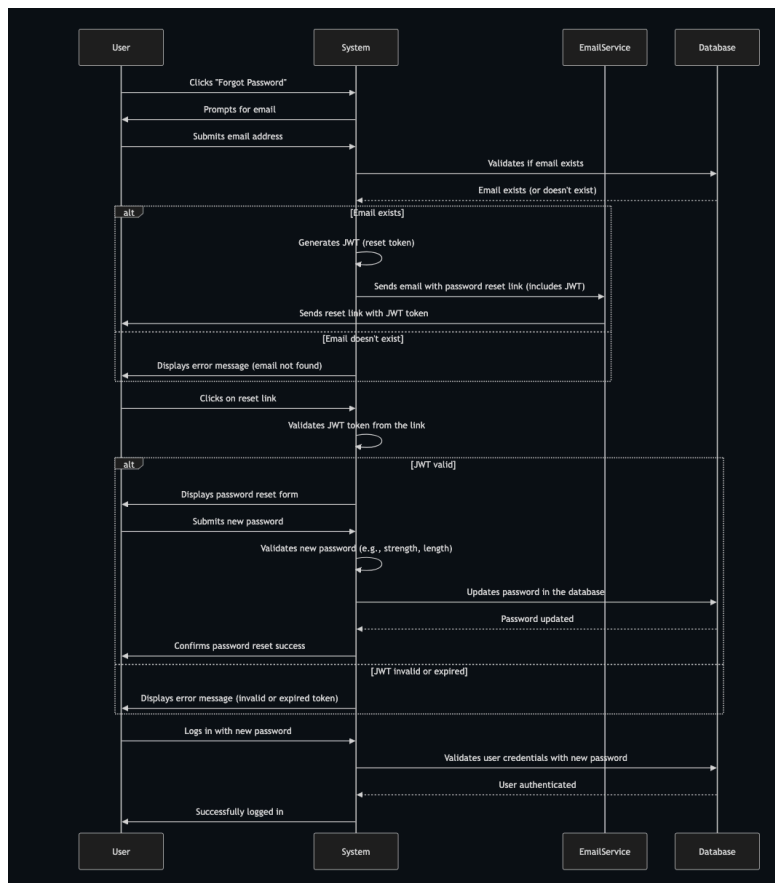


Figure 2. sequence diagram for forgotPassword process

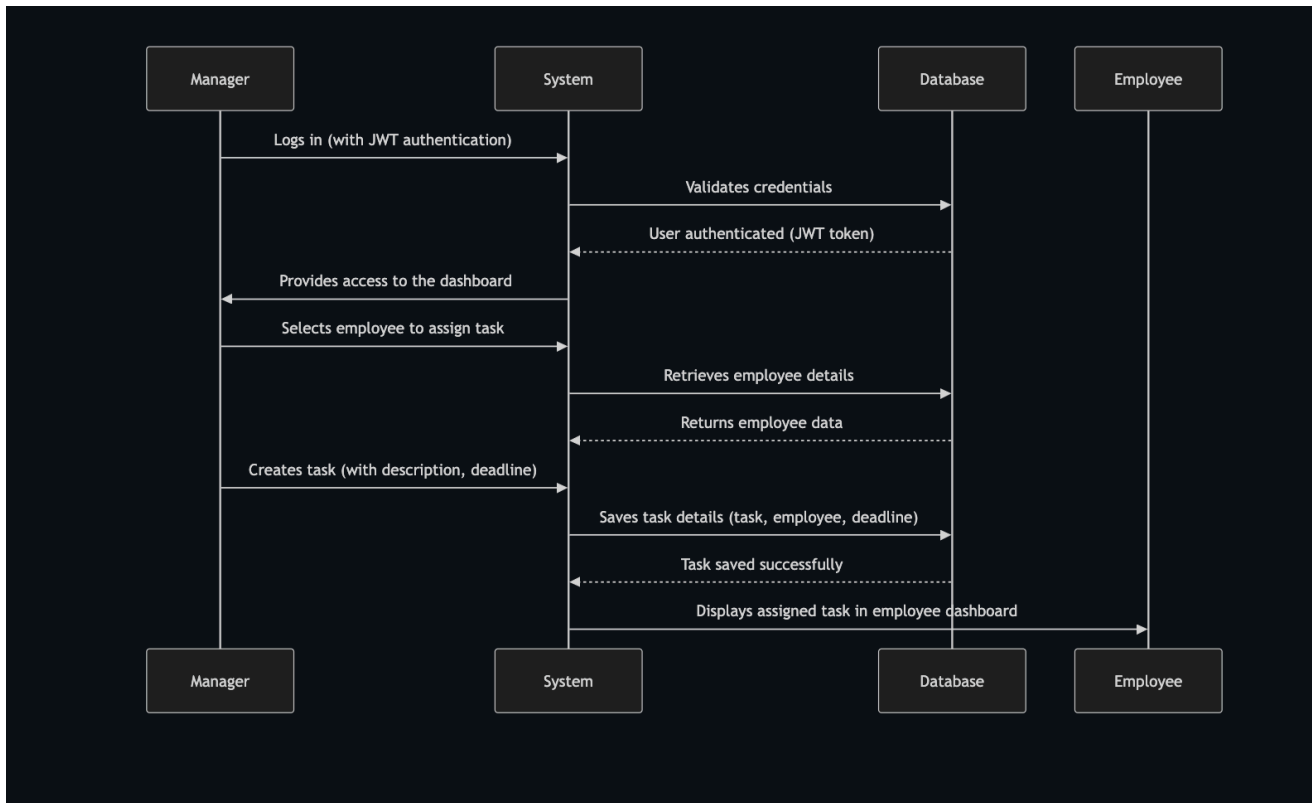


Figure 3. sequence diagram for manager assigning task to employee

IV. Class Diagram

This diagram will provide us with a structured overview of the various entities and their relationships within the system, thus contributing to our overall understanding of the project's structure.

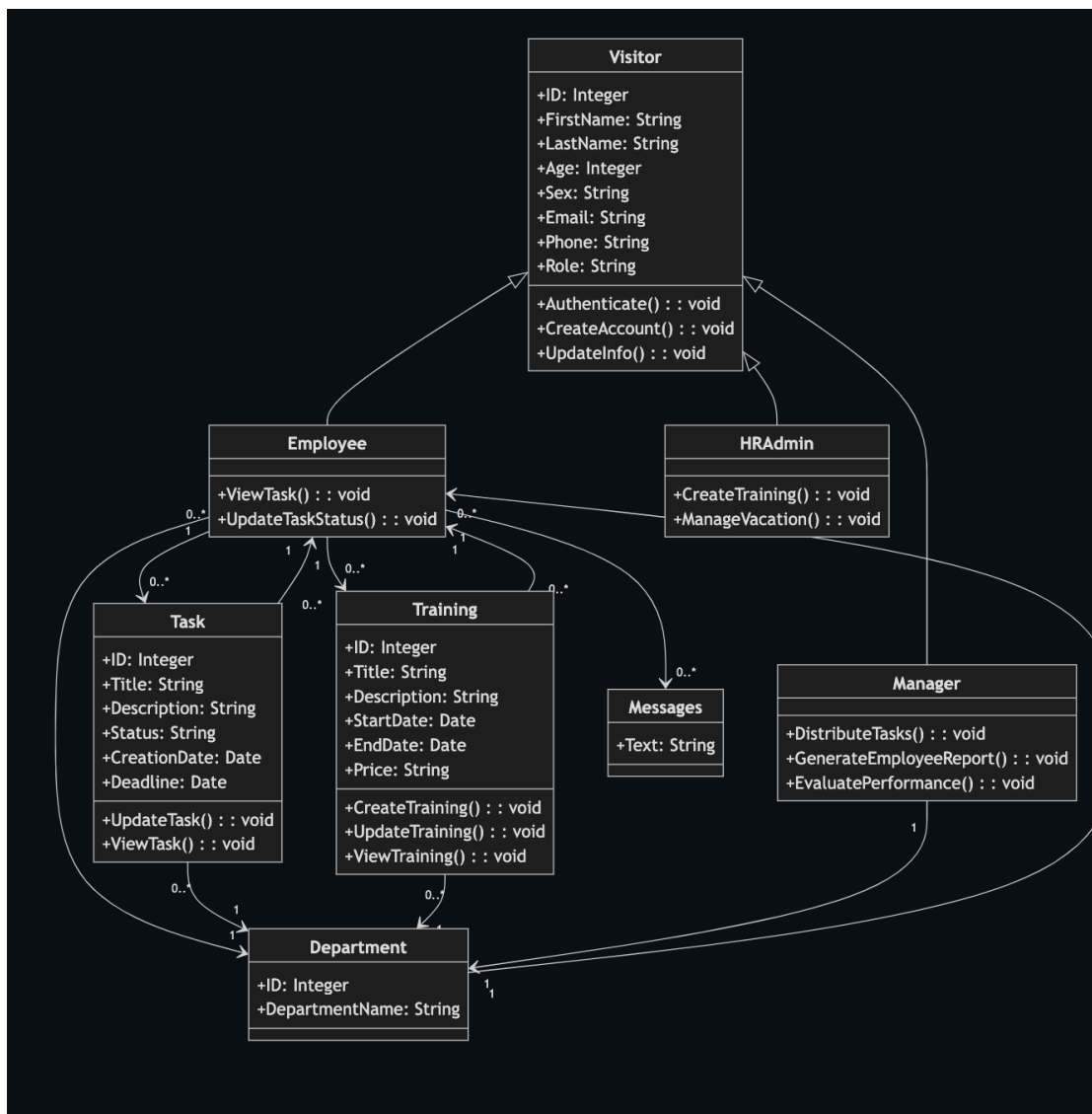


Figure 5. class Diagram

V. Conclusion

In summary, we have explored the key elements of this first section. We examined the interactions through the Use Case Diagram, followed the chronological flow with the Sequence Diagram, and observed the overall structure with the Class Diagram. These key elements provide us with a thorough overview of the project and prepare us for a deeper analysis.

Chapter 3 : Realisation

I. Introduction :

In this chapter, we dive into the crucial phase of realizing our project. After defining the requirements and specifications in the previous chapters, we now explore the complex aspects that contribute to the creation of our system. The realization encompasses essential elements, from the development environment to user interfaces, as well as sequence and class diagrams. This chapter provides a comprehensive overview of the project's execution, focusing on technological choices, development tools, and a detailed description of the key user interfaces that play a central role in the user experience. It represents a fundamental step in the realization of our project, and we are excited to guide you through its complexities.

II. Development Environment

In this section titled 'Development Environment,' we delve into the crucial aspects related to the hardware, software, and programming languages that support our project.

II.1. Hardware Environment

- Model: MacBook Pro 2015
- CPU: Intel Core i7
- Operating System: macOS Monterey 12
- RAM: 16GB
- Storage: 256GB SSD

II.2. Software Environment

In our project, we used Visual Studio Code, a source code editor developed by Microsoft, widely appreciated for its versatility and rich features. This tool allowed us to work efficiently on web development by providing a user-friendly interface, support for many programming languages, debugging features, intelligent autocompletion, and numerous useful extensions created by the community. With Visual Studio Code, we were able to manage our project effectively and produce high-quality code. Additionally, I used Docker to run a PostgreSQL container, enabling an isolated and consistent environment for database management. For working with the database, I used DataGrip, a powerful database management tool. To ensure the API functions correctly, I used Postman for testing the endpoints. Version control was handled through Git and GitHub, enabling seamless collaboration and code management throughout the development process.



Figure 6. VS Code Logo

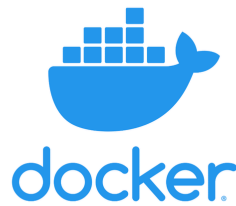


Figure 7. Docker Logo



DataGrip

Figure 8. Datagrip Logo



Figure 9. Postman Logo



Figure 10. Git and Github Logo

II.3. Framework,libraries and design pattern used:

In our project, we used the PERN stack. The PERN stack is a modern and powerful web development stack consisting of PostgreSQL, Express.js, React, and Node.js. It is widely recognized for its efficiency and scalability in building dynamic web applications. This stack provides a complete ecosystem for creating robust web applications, including essential tools such as routing, authentication, and database management. While the stack relies on JavaScript for both client and server-side development, I used the MUI library for React to streamline development and improve the user interface design. Additionally, Prisma was used as an ORM (Object Relational Mapping) to handle database operations efficiently and seamlessly integrate PostgreSQL with the application. This combination of technologies made the project highly versatile and ideal for scalable web applications.

Now, let's dive deep into the frameworks and libraries used, along with the languages they are associated with.

React

A JavaScript library for building user interfaces, primarily for single-page applications. It allows developers to create reusable components and efficiently update the UI using a virtual DOM. React's declarative syntax and component-based architecture make it scalable and maintainable. To streamline development and improve user interface design, we used MUI (Material-UI), a popular React component library that provides pre-designed, customizable UI components following Material Design principles.

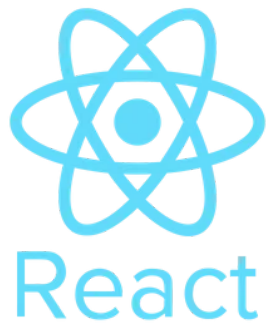


Figure 11 . React Logo



Figure 12 . Material ui logo

Express.js

A fast, minimal, and flexible Node.js web application framework that simplifies the development of server-side applications. It provides a robust set of features for building APIs, handling HTTP requests, routing, middleware, and managing server-side logic. Express.js is widely used for creating scalable and efficient backend applications with Node.js.



Figure 13 . ExpressJS logo



Figure 14 . NodeJs Logo

Prisma :

is an ORM (Object-Relational Mapping) tool for Node.js and TypeScript that simplifies database interactions. It provides an easy-to-use API to query, insert, update, and delete data from relational databases like PostgreSQL, helping developers work with databases more efficiently while maintaining type safety and performance.



Figure 15 . Prisma Logo

MVC (Model-View-Controller) is a software architectural pattern used to separate concerns in an application. The Model represents the data and business logic, the View displays the data to the user, and the Controller handles user input, updates the Model, and refreshes the View accordingly. This separation allows for easier maintenance and scalability.

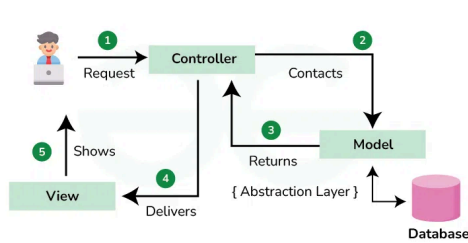
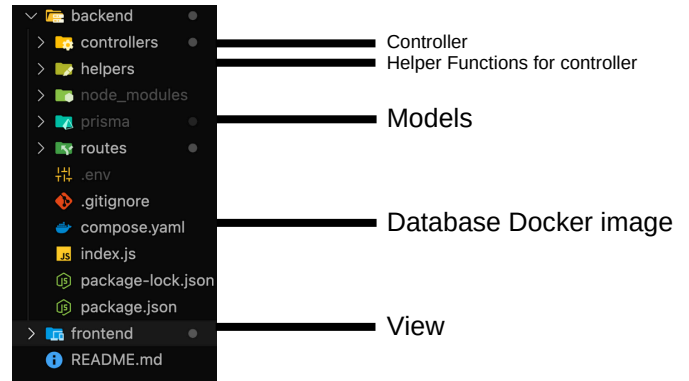


Figure 16 . Mvc chart



III. User interface Description

The login interface features a 'TECH HR' logo at the top. Below it is a 'Welcome to Login Section' heading. The login form includes an 'Email' input field, a 'Password' input field, and a blue 'Login' button. At the bottom, there are links for 'Don't have an account? Register Now' and 'Forgot your password?'.

Figure 17. Login interface



Welcome to Registration Section

First Name

Last Name

Age

Phone Number

Sex

Mobile

Email

Password

Signup as

Department

Select Department

[Sign Up](#)

[You have already an account ? Login Now](#)

Figure 18. *Signup interface*

Forgot Password

[Send Reset Link](#)

Figure 19. *Forget Password interface*

Employee Interfaces

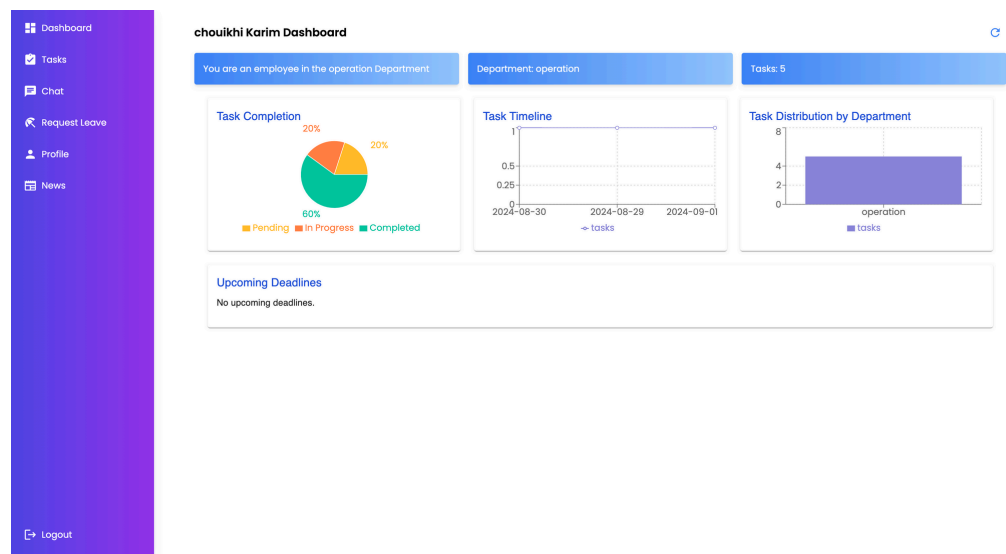


Figure 20. Employee dashboard interface

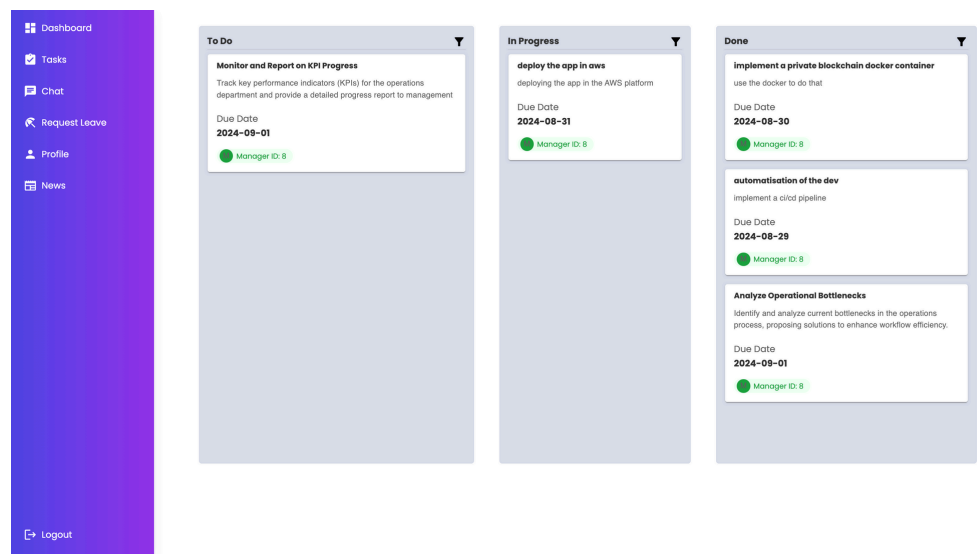


Figure 21. Employee Tasks interface

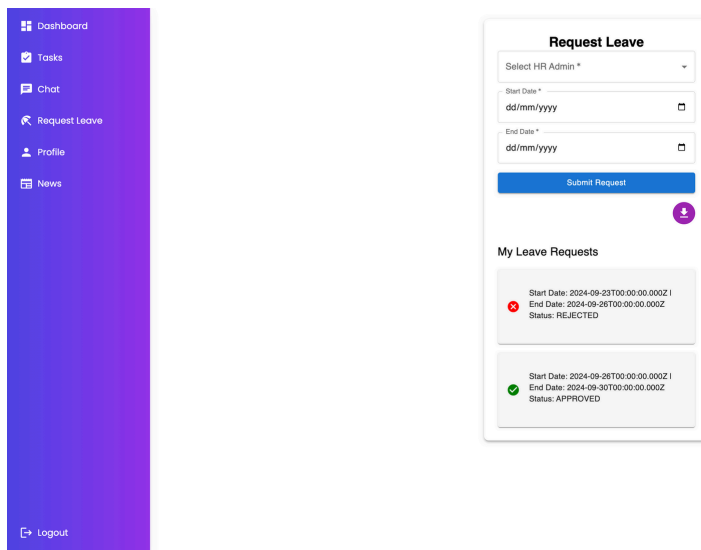


Figure 22. *Employee Request Leave interface*

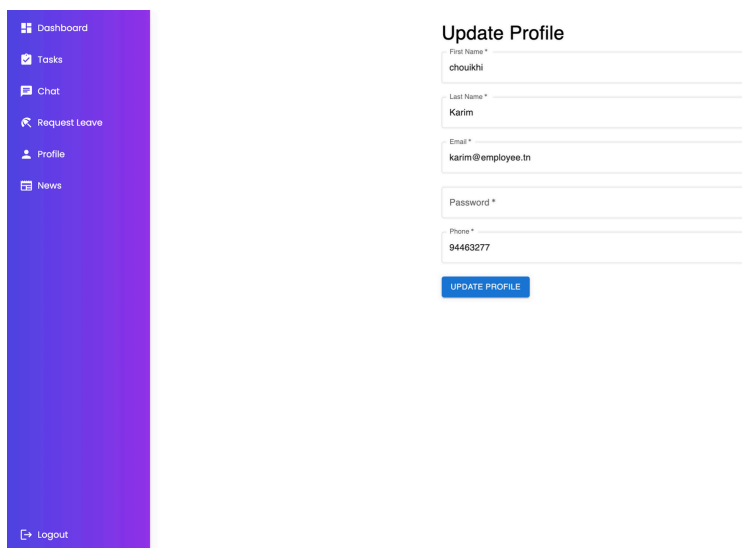


Figure 23. *Employee Profile interface*

Manager Interfaces

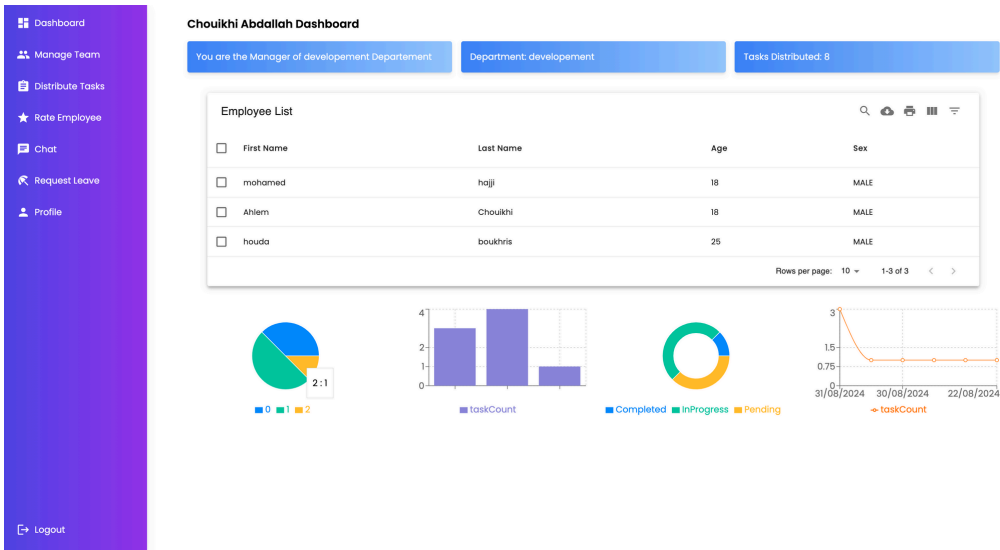


Figure 24. Manager dashboard Interface

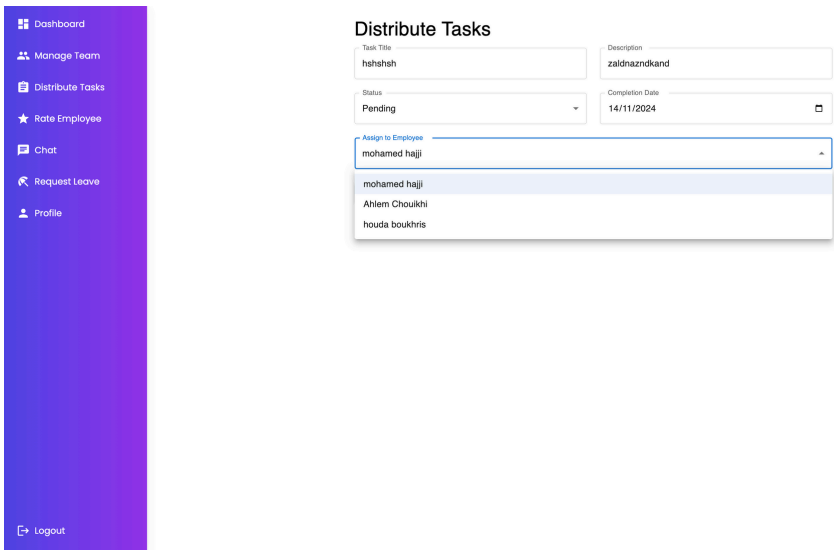


Figure 25. Manager Task Assignment Interface

Hr interfaces

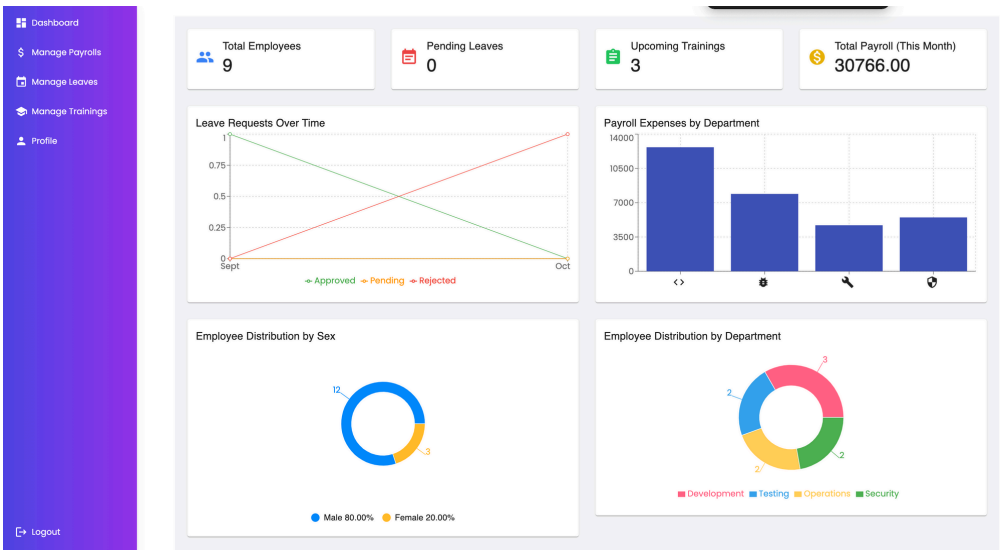


Figure 26. Hr dashboard interface

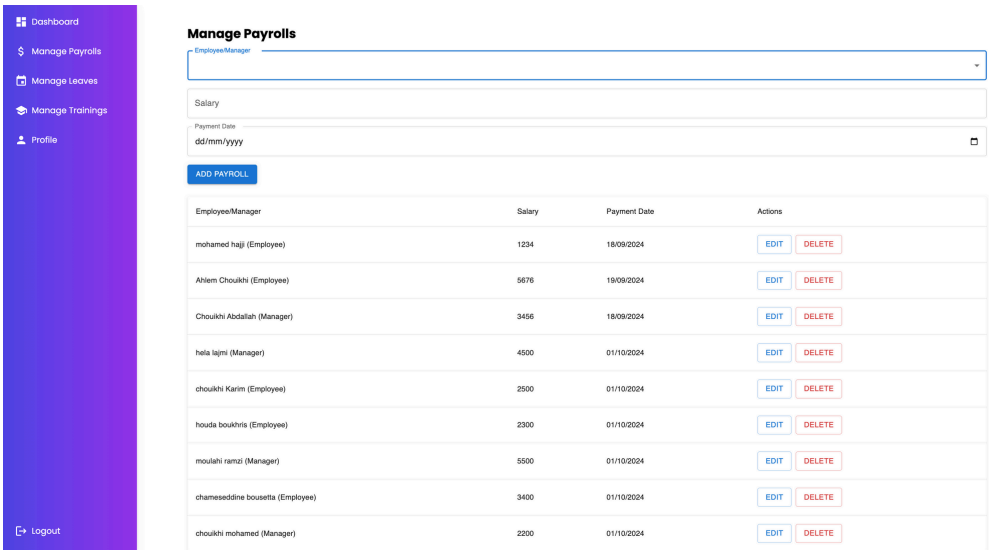


Figure 27. Hr Payroll interface

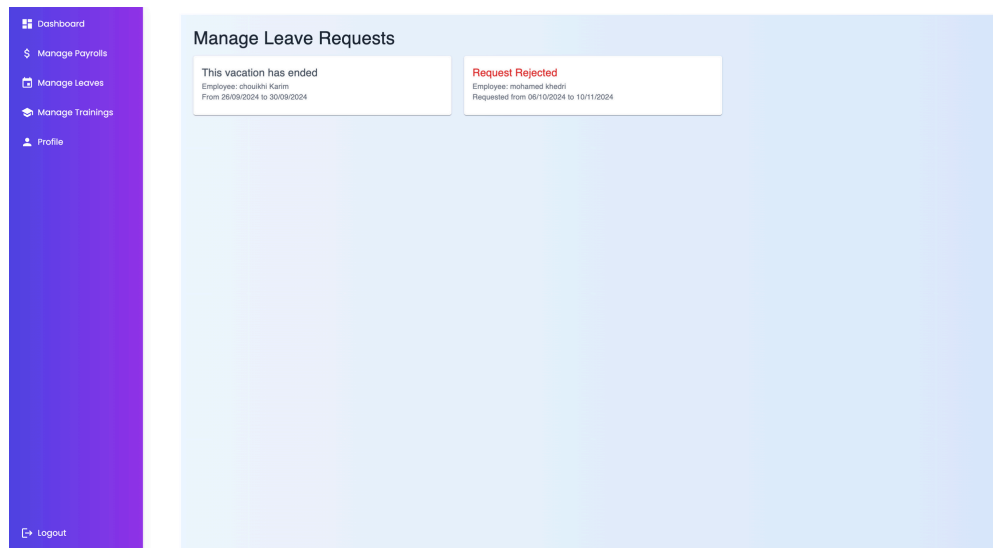


Figure 28. hr Leave Request interface

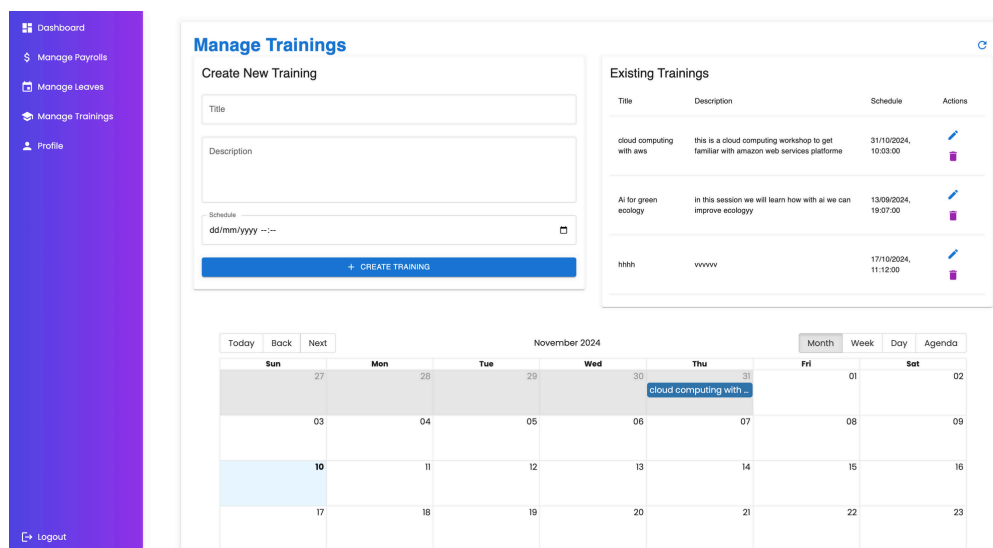


Figure 29. hr Training interface

Genral Conclusion

In conclusion of my internship at Elitech Consulting, I would like to express my sincere gratitude to the entire team for giving me the opportunity to learn and grow within this dynamic company. My experience over the past few months has been exceptional, providing me with a deep understanding of web development, especially through the HR management system project I worked on.

During this time, I had the privilege of gaining practical expertise in using the PERN stack (PostgreSQL, Express.js, React, Node.js), a modern and powerful tool for building dynamic web applications. I was able to apply these newly acquired skills to design and implement a sophisticated HR management system. This project was an invaluable learning experience, allowing me to enhance my skills in backend development, user interface design, and database management.

I also had the chance to collaborate with a talented and passionate team that guided me throughout the process, sharing their knowledge and expertise. This teamwork experience demonstrated the importance of communication, coordination, and idea sharing to achieve common goals.

My time at Elitech Consulting also helped me refine my career goals and focus on a future in web development. I leave this company with increased confidence in my skills and a clear vision for my professional future.

I want to extend my deepest appreciation to the Elitech Consulting team for their constant support and dedication to my success. My time here has been incredibly enriching, and I am eager to apply everything I have learned in my future career.